EPSR-test-ex92 Exponential core and container force with gradient limit

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises the gradient limit (second value of `expecf` > 0) of the default exponential repulsive core and of the exponential container force in the `epsr` engine of EPSR [1,2]: beyond the limit the exponential is continued linearly so that the energy of strongly overlapping or escaped atoms cannot explode [3].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `generate_rp` (exponential core with `gradlimit`, `loggradlimit`), `repulsivept`
- `containerforces_mod.f90`: `en_containerforce` (exponential branch), `get_containerforce_list`
- `ato_manipulation_rev4.f`: `fmolec1`, `atogen`; `molfit.f`: `en_change`; `read_and_save_files.f`: `read_pcof`

Rust (port, rust/epsr-core/src/):

- `potential.rs`: `generate_rp`, `repulsivept`
- `containerforces.rs`: `en_containerforce`, `get_containerforce_list`
- `energy.rs`: `fmolec1`, `en_change`; `structure.rs`: `atogen`; `pcof.rs`: `read_pcof`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. As in EPSR-test-ex15 every DMSO carries an extra massless, uncharged Q site at its sulphur (3400 atoms) and water oxygens further than 4 Å from the nearest Q site are pulled back by an exponential container force. With `expecf 0.300 1.500` the core pcoefex·exp(ddiff) becomes pcoefex·gradlimit·(1 + ddiff - log(gradlimit)) for ddiff >= log(gradlimit), and the container force is limited in the same way.

Inputs

`scripts/make_ex15.py ../../../EPSR-test . 4.0 0.5` (as EPSR-test-ex15: Q sites, 28 pairs, Q-O1 `rmaxex 4.0`, `pcoefexmax 0.5`, `iinit 1`), then `scripts/make_case.py`: `expecf 0.300 1.500` and the repulsive-core lines of EPSR-test-ex1 (`rminex pcoefex coordex`: S-S `4.0 0 0.1`, O1-O1 `2.6 0.5 0`, O1-H1 `1.5 0.2 0.05`).

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
The README reports a gcov check of the Fortran: `generate_rp` executes both the linear and the `exp(ddiff)` term of the core with `gradlimit` = `1.5`, and `en_containerforce` its exponential branch with 345 linear and 2055 exponential evaluations. EPSR-test-ex15 covers only the default `gradlimit` = `1`, `loggradlimit` = `0` of the container force and no exponential core, so this case adds both limited branches.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.